11-4 Queues

Point!

Queues

([1]**Queue**) : A line formed when customers wait for a service, such as at a supermarket register or an attraction at a theme park.

<Example> Queue at the register of a supermarket

[1] Service start time

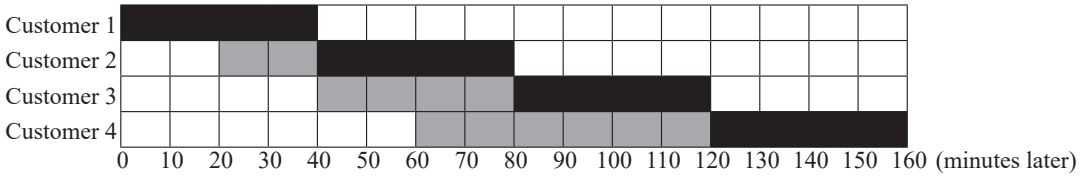
- When there are no customers at the register (no queue)
  Service start time = Time when a customer ([2]**arrives**) at the register
- When there is a previous customer at the register (when there is a queue)
  Service start time = Time when the previous customer ([3]**finished**) getting service

[2] Service end time = Service start time + ([4]**Service duration**)

[3] Waiting time = Service start time - ([5]**Arrival time**)

Warm Up

The following diagram represents changes in the queue at a store, with the vertical axis indicating the number of customers in line and the horizontal axis representing the time in minutes since the arrival of customer 1. The time each customer was receiving service is shaded in black, and the time they were waiting in line is shaded in gray. Answer the following questions.



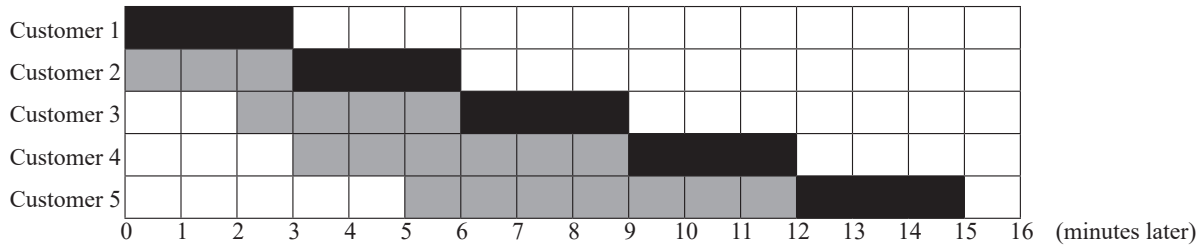
(1) How many minutes did it take to provide service to one customer at this store?
(2) How many minutes after customer 1's arrival was customer 3 able to receive service?
(3) How many minutes did it take from when customer 4 arrived in line until they finished receiving service?
(4) Among customers 1 to 4, who had the longest waiting time?
(5) For how many minutes was the queue the longest?

Explanation

(1) The black cells represent the time when service was being received; therefore, the answer is 40 minutes.
(2) When calculating the time between the arrival of customer 1 and the time customer 3 was able to receive service, 80 minutes have elapsed.
(3) Since customer 4 started waiting in line 60 minutes later and finished receiving service 160 minutes later, the time it took from when customer 4 arrived in line until they finished receiving service is 100 minutes.
(4) From the chart, the waiting times for each customer were as follows: customer 1 was 0 minutes, customer 2 was 20 minutes, customer 3 was 40 minutes, and customer 4 was 60 minutes. Therefore, the customer with the longest waiting time was customer 4.
(5) The queue was longest during the 20-minute period from 60 minutes to 80 minutes, when customers 2, 3, and 4 were waiting in line.

Try

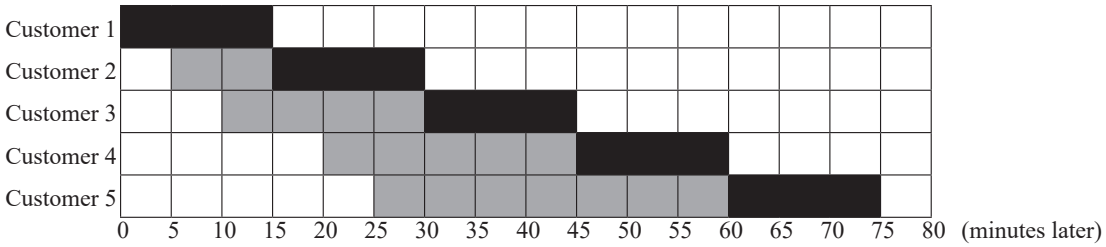
The following diagram represents changes in the line at reception, with the vertical axis indicating the number of customers in line and the horizontal axis representing the time in minutes since the arrival of customer 1. The time each customer was receiving service is shaded in black, and the time they were waiting in line is shaded in gray. Answer the following questions.



(1) What is the term for a line that forms at reception desks, registers, attractions, etc., where customers line up in order to receive services?
(2) How many minutes was the wait time for customer 2?
(3) How many minutes after the arrival of customer 1 did customer 3 arrive in the queue?
(4) How many minutes after the arrival time of customer 1 did customer 4 finish receiving service?
(5) Among customers 1 to 5, who had the longest waiting time?
(6) What was the maximum length of the queue between the arrival of customer 1 and the completion of service for customer 5?

Exercise

1 The following diagram represents changes in the queue at a store, with the vertical axis indicating the number of customers in line and the horizontal axis representing the time in minutes since the arrival of customer 1. The time each customer was receiving service is shaded in black, and the time they were waiting in line is shaded in gray. Answer the following questions.



(1) How many minutes is the service time per customer at this store?
(2) How many minutes after customer 1's arrival was customer 4 able to receive service?
(3) How many minutes did it take from when customer 5 arrived in line until they finished receiving service?
(4) Among customers 1 to 5, who had the longest waiting time?
(5) For how many minutes was the queue the longest?